

Precision biomarkers of the exercise-induced immune “open window”: a scoping review of the validation readiness of conventional and omics candidates

Protocol version: 3.0 (draft for team freeze) — 4 October 2026

Status: Pre-archive working protocol; formal searches, screening and synthesis have not begun.

Target journal: *Frontiers in Immunology*, under the Systematic Review article type, which includes scoping reviews; specialty section to be selected. Submission will include the PRISMA-ScR checklist and flow diagram as supplementary material and an AI-use disclosure in both Methods and Acknowledgments stating tool name, version, model and source.

Registration and archiving: Public GitHub repository with a GitHub Release archived by Zenodo, which assigns a dated, immutable version DOI and a concept DOI on release. No release exists, no DOI has been assigned, and this project has never been registered in any registry; PROSPERO does not accept scoping reviews and OSF Registries is no longer the planned route.

Team: Access to all five databases and a second human reviewer are confirmed. Team members are identified by codes: A (principal investigator, independent PRESS reviewer, analysis and writing lead), B and C (independent reviewers), D (search lead, data manager, adjudicator). Funding and competing-interest declarations remain to be supplied.

This is the canonical English protocol. The Chinese protocol is an aligned implementation version. The rendered English source and PDF are generated from this complete document; they are not a shortened independent protocol. Operational appendices in `03_search-06_synthesis` implement the decisions below. This version follows three pre-archive activities disclosed in J: an internal protocol (version 1.0, January 2026), public exploratory reconnaissance and source verification on 2 October 2026 (Codex), and a collision re-check of the narrowed framing against published and registered reviews on 4 October 2026 (Claude Code; `02_preliminary/collision_recheck_2026-10-04.md`). The protocol date is not a completed formal search cutoff. No search yield, eligibility decision, included-study count or review result is claimed here.

A. Background and rationale

Exercise can alter circulating immune-cell counts, immune mediators, mucosal measurements and immune-cell function over different time scales. The “open window” has been used to describe the hours to days after strenuous or prolonged exercise during which immune changes have been proposed to coincide with reduced host protection [15,16]. Whether such a period exists remains debated: lymphocytopenia may reflect redistribution rather than loss of whole-body function, and plasma-volume change, circadian timing and sampling alter interpretation [1,2]. This review does not adjudicate that debate. It uses the open window solely as an organizing construct that defines the post-exercise period (0–72 hours after cessation of an identifiable bout), the sampling time points and the marker families within which candidates are charted, irrespective of whether original authors endorse, reject or omit the hypothesis.

Candidates are charted by marker family. Conventional families include leukocyte and lymphocyte redistribution, natural-killer-cell counts and cytotoxicity, neutrophil function, T-cell subsets and function, mucosal secretory IgA, cytokines and other inflammatory mediators, and immunoglobulins and complement. Omics families include immune-cell transcriptomics, proteomics, metabolomics and lipidomics, epigenomics, non-coding RNA and single-cell measurements. Higher measurement dimensionality does not alone confer immune specificity or individual predictive value. A group-average response, an in-silico pathway annotation, an association with symptoms and a validated individual prediction address different questions. Candidate usefulness also depends on the exercise exposure, tissue or fluid, analytical method, sampling interval and intended monitoring use.

The intended contribution is a per-marker **validation-readiness map** of candidates measured within the organizing window. Throughout this protocol and its appendices, a validation-readiness map means a descriptive, domain-by-domain profile, for each marker or marker family, of which of the seven precision-evidence domains defined in C have actually been investigated and with what result. It is not a score, ranking, grade or development ladder, and every later prohibition on readiness scores or rankings refers to this definition. The map emphasizes individual baselines, within-person variation, repeatability, time-resolved interpretation, functional or clinical links and independent validation. It is not a catalogue that labels every statistically significant omics feature a precise biomarker. Analytical and clinical validation must be interpreted for a specified context of use [3].

In their 2017 review of immune recovery after exercise, Peake and colleagues concluded that, “except for salivary IgA, clear and consistent markers of this immunodepression remain elusive” [15]; that statement is the auditable starting point for this map. The debate on whether the open window exists is held by Peake et al. 2017 [15], Campbell and Turner 2018 [1], Simpson et al. 2020 [16], Reitzner and Brodin 2026 [19] and Xie et al. 2026 [17]; this review does not join it. Narrative reviews have also proposed marker agendas around the open window: Xie et al. 2026 in *Frontiers in Immunology* (tissue-resident memory T cells and their circulating precursors) [17]; Shi et al. 2025, also in *Frontiers in Immunology* (workload-dependent recovery of conventional indices) [18]; Reitzner and Brodin 2026 (the immune system as a systems-level biosensor of health and performance) [19]; and Mănescu et al. 2026 (molecular biomarkers of training responses) [20]. Davison et al. 2025 questioned the clinical relevance of ex vivo immune assays [2]. None of these charts human evidence marker by marker against prespecified validation domains with a reproducible search, and systematic reviews to date address single marker families or single populations: exercise-associated non-coding RNAs, B-cell-related immunity, TH1/TH2 markers in older adults and exercise immunometabolomics [4–7]. The registered reviews nearest to this protocol differ in Population–Concept–Context: PROSPERO CRD42020186264 is confined to salivary IgA and respiratory-infection risk; PROSPERO CRD42022324139 is confined to the acute effect of high-intensity interval training; and OSF osf.io/4hsdr shares the JBI and PRISMA-ScR method stack but is restricted to athletes aged 10–17 years and to conventional markers, without omics, open-window or validation dimensions, whereas this review includes adults only. The 4 October 2026 re-check that identified these items is filed at 02_preliminary/collision_recheck_2026-10-04.md. We make no “first review” or “no competing review” claim. The proposed distinction is the joint, time-resolved charting of conventional and omics candidates within one organizing window against separately recorded precision domains. That distinction will be rechecked against published and registered reviews before the v3.0 archive release and at the final search update.

A scoping review is appropriate because the primary questions concern the distribution, measurement, interpretation and validation of heterogeneous evidence, rather than one intervention effect or diagnostic threshold. We will follow JBI scoping-review methods [8], report with PRISMA-ScR [9], and report searches with PRISMA-S [10]. Prespecified, design-appropriate critical appraisal will support interpretation. We will not conduct a meta-analysis or turn appraisal into a clinical ranking.

B. Objective and review questions

This scoping review will map human evidence on candidate immune biomarkers measured within the exercise-induced “open window”, used here solely as an organizing construct that defines the post-exercise period after an identifiable bout, the sampling time points and the marker families to be charted. Across conventional immune indices and omics candidates in generally healthy adults after single identifiable bouts (Core A) and across repeated bouts in the same individuals (Support B), we will chart, marker by marker, which of seven precision-evidence domains have actually been investigated: analytical reliability, temporal validity, immune specificity, individualization,

functional or clinical linkage, independent validation and use, and feasibility. The resulting per-marker validation-readiness map is a descriptive domain-by-domain profile, not a score, ranking or development ladder. The review does not test whether an open window of increased infection susceptibility exists, and original authors’ endorsement or rejection of that hypothesis neither determines eligibility nor weights evidence.

- 1. **Measurements and context:** Which marker families, individual biomarkers, analytical platforms, immune compartments and sample matrices have been measured within the organizing window (0–72 hours after cessation), in which adult populations and exercise settings?
- 2. **Timing, confounding and interpretation:** Which time points within the window (immediate, 30 minutes to <3 hours, 3 to <24 hours, 24 to 72 hours) and which outside-window recovery points are covered; how are cell composition, hydration, sampling and other relevant confounders addressed; and how do original authors interpret their findings in relation to the open window (charted descriptively only)?
- 3. **Individualization:** What evidence exists on personal baselines, within-person variability, repeatability across bouts, response heterogeneity, subgroup interpretation or individualized models, and how has it been validated?
- 4. **Function, outcomes and use:** Which functions or clinical outcomes were merely co-measured, which were associated with biomarkers, and which monitoring or prediction uses underwent independent validation? Together with questions 1–3 this yields the per-marker validation-readiness map and the development requirements that remain.

Positive, null and inconsistent findings are all relevant. The questions do not presuppose that exercise is immunosuppressive, nor test whether an open window exists, nor assume that any candidate is ready to predict infection.

C. Framework and operational definitions

Eligibility uses Population–Concept–Context (PCC). “Post-exercise” begins at cessation of an identifiable bout or competition. Exact times and their reported origin are retained. During-exercise samples can be ancillary observations in an eligible study but cannot alone establish post-exercise eligibility.

The **organizing post-exercise window** is the period from cessation to 72 hours inclusive. It is a charting frame, not an eligibility cap: a study qualifies through at least one post-cessation immune-related sample, and samples taken after 72 hours in an otherwise eligible study are extracted in full and tagged “outside-window recovery” because they document recovery endpoints. No exclusion code refers to the window. **Author open-window interpretation** is a descriptive tag recording whether the original authors used the open-window term, invoked immunosuppression or susceptibility without the term, interpreted changes as redistribution, or offered no window interpretation; it affects neither eligibility nor weighting.

For this project, **precision monitoring** means interpreting an immune-related measurement within the organizing window for a defined person or subgroup, exercise context, sampling time and intended monitoring use. Seven domains are recorded separately:

Domain	Evidence to chart
Analytical reliability	Assay performance, limits, quality control, reproducibility and pre-analytical handling
Temporal validity	Exact exercise-to-sample intervals, position relative to the organizing window, longitudinal coverage and how response or recovery is defined

Domain	Evidence to chart
Immune specificity	Direct immune measurement, cellular origin, composition adjustment and relevant alternative explanations
Individualization	Personal baselines, within-person noise, repeated-bout reproducibility, response heterogeneity, subgroup or individual models
Functional or clinical linkage	Direct immune-function measurements, tested associations and explicitly defined clinical outcomes
Independent validation and use	Technical confirmation versus independent participants/cohorts, model validation and tested decision use
Feasibility	Sampling burden, accessibility, cost, turnaround and field implementation, when reported

These are project operational domains, not a validated scale, regulatory designation or ordinal development ladder. There will be no combined “precision score”; domain evidence is summarized only as the per-marker validation-readiness map defined in A. Omics measurement, machine learning or a group mean difference alone does not establish personalized usefulness. Individualization or clinical validation is **not required for inclusion**; absence of such reporting is itself a mapping finding.

A **response** is an observed post-exercise measurement or contrast. A **recovery trajectory** requires longitudinal observations and an explicit endpoint; a single early sample or a nonsignificant comparison with baseline does not establish recovery. Cell-count normalization, biomarker normalization, immune-function recovery and clinical recovery remain distinct.

Immune linkage requires a direct immune-cell/count/subset/effector measure; a recognized immune mediator such as an immunoglobulin, cytokine or complement component; omics measured in immune cells; or biofluid omics with an immune objective or analysis stated in the original aims/methods and identifiable immune candidates. Original-study preregistration is not required. Mere post-hoc immune enrichment in an otherwise general muscle or metabolic study is insufficient for the main map. Cortisol, creatine kinase, lactate or heart-rate variability alone do not qualify, but may be charted as covariates in otherwise eligible studies.

Direct functional assays are distinguished from computational annotation. Co-measurement of a function or clinical outcome does not establish a biomarker–outcome association. Self-reported symptoms, clinician-defined syndromes and pathogen-confirmed infection are separate outcome classes. Association, temporal prediction, calibration, discrimination and clinical utility are recorded separately. We use “prediction” for future outcome prediction here, without assuming the treatment-response meaning of a predictive biomarker in other taxonomies.

Missing information is coded **NR** (not reported), **NA** (not applicable) or **UNCLEAR** (information present but insufficient or ambiguous after checking). NR is not evidence of biological absence or methodological failure.

D. Eligibility criteria

D.1 Population

Eligible participants are adults aged 18 years or older, of any sex or training status, who are generally healthy at baseline and do not have an active infection. Athletes, recreational exercisers and untrained participants are eligible; training strata are charted separately. Older adults are eligible and flagged as a subgroup. Children and adolescents (<18 years) are excluded; mixed-age cohorts are eligible only where an adult stratum is separable in the report. This boundary distinguishes the review from the registered youth-athlete scoping review osf.io/4hsdr (10–17 years). Extractable healthy adult strata within cohorts of mixed health status are also eligible. Cohorts with known clinical/metabolic disease or disease-treatment exercise programmes are excluded from the main map. Ambiguous metabolic-health descriptions, including insulin resistance without a clear clinical classification, require explicit adjudication using the reported diagnosis and separability of strata, rather than an automatic healthy designation.

D.2 Exercise context and evidence streams

Core A — acute challenge: An identifiable single bout or competition of endurance, resistance, interval/sprint or mixed exercise, with at least one post-cessation immune-related sample and either a pre-bout baseline or a suitable non-exercise/comparator condition. No intensity, duration or mode threshold is imposed: any identifiable bout qualifies, and intensity, duration and mode are recorded as reported and used as evidence-map strata, never as eligibility filters, because the open-window literature defines “prolonged” and “intense” inconsistently and a threshold would discard comparator evidence. Immediate samples are eligible and there is no minimum 30-minute follow-up. The organizing window of 0–72 hours after cessation frames the charting; it is not a maximum interval, and samples beyond 72 hours in an eligible study are extracted and tagged “outside-window recovery” (C). During-only studies are ineligible. Repeated acute challenges before and after a training programme qualify for A; training moderation is extracted separately. Immediate-only studies contribute response evidence, not a complete recovery trajectory.

Support B — repeated post-exercise windows in the same people: The same immune-related marker is measured in the same individuals in the post-exercise windows after at least two identifiable bouts or competitions, with a known or reliably recoverable exercise-to-sampling relation and a baseline or appropriate comparator enabling interpretation. B is the cross-bout monitoring layer and the main evidence source for the individualization and repeatability domains; it is presented separately from A. A cohort may meet both A and B; linked cohort identifiers prevent duplicate overall counting. General weekly or seasonal resting surveillance with unknown last-bout timing, training pre/post snapshots with unknown exercise-to-sampling intervals, and habitual-exercise cross-sectional comparisons are background only. A long observation period cannot substitute for an identifiable post-exercise time relation.

Studies involving supplementation, diet or other co-interventions qualify only where an exercise contrast is interpretable separately, for example through an appropriate control/placebo arm or factorial contrast. Otherwise the inseparable intervention is ineligible. Prior training, co-interventions and recovery arrangements are extracted.

D.3 Concept, design and publication status

Eligible studies meet the immune-link definition in C and report original human measurements. There is no requirement for the words “precision,” “biomarker,” “open window” or “recovery,” for a particular direction or significance level, or for a functional/infection endpoint. Studies are neither selected nor weighted according to whether their authors endorse, reject or ignore the open-window hypothesis; the author interpretation is charted as a descriptive tag (C). Designs may include randomized or nonrandomized experiments, crossover studies, before–after studies

and observational designs that satisfy the timing and comparator rules. No arbitrary minimum sample size is imposed; small designs are clearly identified and appraised appropriately.

The primary A/B map includes peer-reviewed original reports. Reviews support background and citation chasing. Animal-only or in-vitro-only studies, protocols, editorials, conference abstracts without a full original report and pure methods papers without eligible human exercise observations are excluded from the primary map; potentially relevant abstracts are tracked for full publications.

A bounded supplementary search of bioRxiv and medRxiv will identify publicly available preprints meeting the same substantive PCC criteria. They enter a separately labelled frontier register, with separate flow counts and denominators, and are not pooled with peer-reviewed evidence. Publication status is checked at the final update, and preprint/published versions are linked and merged appropriately. Clearly labelled public preprints may be cited under current Frontiers guidance [14]; this does not imply access to unpublished private results.

D.4 Date, language and retrieval

Searches run from database inception to their actual execution dates. No search language restriction is applied. English and Chinese full texts are read directly; other languages use translation with independent human checking of eligibility and key data. Unresolved language or eligibility information remains awaiting classification rather than being automatically scientifically excluded.

Institutional access and lawful public copies will be used for full texts. A sought but unobtained report is “not retrieved,” not a scientific full-text exclusion. This protocol does not plan author outreach; no contact is sent without separate authorization. Retrieval attempts and unresolved reports are logged.

E. Information sources and search strategy

The five planned databases are PubMed, Web of Science Core Collection, Scopus, Embase and SPORTDiscus. PubMed is reported as PubMed rather than implying a separate MEDLINE-platform search. Embase platform choice is unresolved: draft Embase.com and Ovid translations are supplied for confirmation; execution of one platform is not claimed as two databases. The SPORTDiscus draft uses EBSCOhost. Exact collections, platforms, interfaces and coverage will be recorded.

The planned search is **E AND (I OR O) AND T**, where E represents exercise, I immune concepts, O omics and T post-exercise or acute-bout wording; the immune and omics routes are run within this structure and deduplicated. Controlled vocabulary and free-text fields are adapted to each platform. The immune route covers immune cells, mediators, function and mucosal immune terms; infection outcomes are not search concepts, because an infection outcome alone does not meet the immune-link rule (C), and infection outcomes are charted whenever an eligible study reports them. The omics route covers transcriptomics, RNA/non-coding RNA, proteomics, metabolomics, lipidomics, epigenetics and multi-omics. The exercise block includes relevant training, competition and sport terminology; the broad **train*** truncation is retained until its marginal yield has been tested on a pilot sample. The search is by construct, not by label: “precision,” “biomarker,” “open window” and “recovery” are not mandatory search concepts, and the organizing window and all eligibility rules are applied at screening rather than in the query. The phrase “open window” is unsuitable as a filter—a PubMed title/abstract query combining it with exercise and immune terms returned 18 records for 2018–2026 at the 4 October 2026 re-check, and original studies rarely use it—so open-window-type terms may be added only as OR supplements or used for citation chasing, never as AND terms.

Amendment PRE-003 (4 October 2026) makes T a required concept instead of a diagnostic block;

T expresses that measurements follow an identifiable bout and does not encode the 0–72 hour window, which is still applied at screening. In PubMed parser validation on 4 October 2026, the construct-at-screening query E AND (I OR O) returned 255,019 records, about 8,500 person-hours of two-reviewer title/abstract screening before records from the other four databases (assumed to add 1.5–2.5 times the PubMed volume before deduplication), whereas requiring T returned about 15,900 records in the decision test and 20,090 records with the finalised v0.7 T block, retrieving all 10 positive PubMed seeds in both runs. That evidence is limited: only 10 positive seeds were tested, the proximity form that recovered one missed seed was designed after the miss, and T is expected to miss reports that give timing only numerically (for example “at 1 h and 24 h”), so the trade-off is disclosed as recall given up for feasibility. Mitigations are a positive seed set expanded to at least 30 reports, including at least five that give timing only numerically, re-tested in the pilot; a T-free sensitivity run during the pilot, in which a random sample of records retrieved only without T is screened; and backward and forward citation chasing of all included reports and of the five open-window reviews named as citation-chasing sources (Peake et al. [15], Campbell and Turner [1], Simpson et al. [16], Xie et al. [17] and Shi et al. [18]).

Full working strategies and a search-log template are in `03_search`. The PubMed strings were parser-validated on 4 October 2026, and the independent PRESS review [11] was accepted with conditions on the same date (T required, no infection-outcome terms, `train*` retained pending the pilot test); the translations for the four licensed platforms remain **drafts, not platform-validated searches**. PRESS revisions and the resolution of comments are retained. Known eligible and boundary examples will test strategy behavior; the v3 seed list includes open-window-centred original studies of conventional indices (for example salivary IgA, natural-killer-cell and lymphocyte redistribution responses after prolonged exercise) and boundary cases (immediate-only sampling, sampling only beyond 72 hours, youth cohorts and resting training studies). Retrieval of a few selected positive seeds is a debugging check, not an estimate of overall recall. A seed not retrieved will be investigated; strategy changes must retain a conceptual rationale rather than simply fitting known papers.

Backward references and forward citations of included reports and relevant neighboring reviews will be checked, including forward citation tracking of Peake et al. [15] and Campbell and Turner [1] before submission. Registered and ongoing reviews in OSF Registries, PROSPERO, INPLASY, Research Registry and Zenodo, together with adjacent recent reviews, will inform the novelty assessment; the 4 October 2026 registry scan is pre-archive reconnaissance and will be repeated before submission. Supplemental preprint searches will record sites, exact queries, dates, filters, results examined and stopping rules. Citation chasing has a defined sweep and documented closure/update; it is not silently unlimited browsing.

For every run, preserve the exact query, database/platform, run date and time, limits, hits, export counts, file format, immutable raw export and checksum. Record search amendments, supplementary routes and deduplication decisions under PRISMA-S. The target is to update searches within 30 days before submission; this is a project target, not a journal rule. New exports do not overwrite earlier exports.

Records are deduplicated using identifiers followed by normalized title/author/year checks and human confirmation of uncertain matches. Distinct reports of the same study or cohort remain linked, rather than being discarded as bibliographic duplicates. Study, report, cohort and preprint-version relationships are preserved. No formal yields are available at protocol preparation.

F. Selection and calibration

Two human reviewers independently screen **all** titles/abstracts and **all** sought full texts. At title/abstract stage, missing timing or methodological detail normally proceeds to full text rather than being guessed. AI does not make automatic exclusions or replace either reviewer.

After the v3.0 protocol has been archived (J) and before formal screening, a saved random procedure selects 50 records from a frozen deduplicated draft-search pool; the pool is generated after the archive release and retained with its version. The pool version, eligible sampling universe, random seed, selected IDs and reviewer decisions are retained. The project criterion is at least 80% raw agreement plus resolution of all conceptual disagreements; this threshold is not attributed to JBI. Cohen's kappa is reported descriptively with prevalence limitations. If the criterion is unmet, the manual is clarified and a fresh pilot is performed. If the pilot changes an eligibility rule, the protocol is re-released with justification before formal searching (J). The 50-record pilot does not confer final study eligibility. Full-text boundary exercises follow before formal full-text screening.

Disagreements are discussed; unresolved decisions go to a predesignated adjudicator whose name must be supplied before formal screening. Changes to an eligibility interpretation are dated and applied to previously screened records when relevant. Operational records retain separate independent decisions and the consensus decision.

The screening manual defines this fixed primary exclusion hierarchy: FT01 ineligible source type; FT02 not original human research; FT03 ineligible population, including cohorts under 18 years without a separable adult stratum; FT04 no qualifying identifiable exercise context, where any intensity, duration or mode qualifies and the code applies only when no identifiable bout, or for B no qualifying repeated-bout relation, can be established; FT05 no post-cessation sample, where sampling only beyond 72 hours is not a reason for this code because the organizing window is a charting frame; FT06 no baseline or suitable comparator; FT07 no eligible immune link; FT08 inseparable mixed intervention. Only the first established scientific reason is assigned at full-text exclusion; secondary concerns and supporting passages remain in notes. Unresolved information is not treated as an established failure. Duplicate reports/versions, not-retrieved reports and awaiting-classification records have administrative statuses outside the scientific exclusion hierarchy. The actual flow will reconcile identification, duplicates, screening, sought/retrieved reports, exclusions and included reports/cohorts, with the frontier register separate. Counts remain unavailable, not zero, until work is performed.

G. Data charting

G.1 Pilot, structure and verification

After the archive release, ten purposively chosen reports will calibrate charting across exercise designs, conventional assays, cell/biofluid omics, early samples, repeated-bout monitoring, and timing and population boundary cases, including an adolescent cohort used as a population boundary case. These reports are training material, not ten included studies, and may include cases ultimately excluded. Their source, selection rationale and unresolved questions are recorded in the pilot manifest. Pilot refinements are versioned before full extraction.

Two humans independently extract eligibility, timing, sample sizes, effects/uncertainty, precision domains, functional/clinical outcomes, validation and appraisal. One may enter purely bibliographic metadata for the other to verify. Discrepancies are resolved against source passages, with adjudication where necessary. The v3 dictionary, template and manual in `05_extraction` define the operational schema; the Excel workbooks used for screening, extraction and appraisal are generated from those JSON templates by `scripts/build_workbooks.py`, and CSV exports are versioned in git. Data use linked report, study and cohort tables with long-form assay/marker/time/comparison records. Raw independent extractions and reconciled data are retained separately.

G.2 Charting domains

- **Provenance:** Identifiers, publication/version status, correction/retraction checks and dates, study/cohort links, source locations, extractor and decision history.

- **Participants:** Eligibility and health definitions, infection screening, recruitment/analysis numbers, age range and separability of the adult stratum, sex, training status, older-adult subgroup, per-group/time/platform participant numbers and within-person pairing.
- **Exposure:** Design, exercise mode, duration, absolute and relative intensity metrics and values as reported, comparator, training background, previous load, co-interventions, environmental conditions and recovery schedule.
- **Sampling:** Exact time and time origin, and position relative to the organizing window (within window, outside-window recovery, pre-exercise baseline or matched control time); matrix/immune compartment; collection, processing, storage, delay/freeze–thaw; circadian timing; plasma-volume/hydration correction; urinary dilution adjustment; salivary flow/secretion rate; cell composition and normalization.
- **Measurements:** Marker family (A); original and standardized marker names/identifiers, modality, targeted/untargeted assay, platform, units, detection/quantification limits, quality control, batch effects, molecular-identification confidence, bulk/single-cell structure and integration method.
- **Results:** Relevant contrasts, direction and magnitude, effect estimates and reported CI/SE or other uncertainty, multiplicity control, missing data, quality-control exclusions and measured/adjusted confounders. Relevant null and discordant findings are retained.
- **Author open-window interpretation:** Whether the report uses the open-window term, invokes immunosuppression or susceptibility without the term, interprets changes as redistribution, or offers no window interpretation, with the source passage; recorded descriptively and never used for eligibility or weighting.
- **Precision domains:** All seven domains in C are charted separately: analytical reliability; temporal validity; immune specificity; individualization; functional/clinical linkage; independent validation and use; and feasibility. Subfields include personal baseline, within-person variation, test–retest/repeated-bout reliability, heterogeneity analyses, subgroup/individual model specification, validation type, participant-level separation, calibration/discrimination, thresholds and decision/implementation evidence where available.
- **Functions and outcomes:** Direct assays versus computed annotation; co-measurement versus tested association; outcome definition/ascertainment, symptom versus clinical versus pathogen outcome, biomarker timing relative to onset and follow-up duration.

For large omics datasets, chart the assay-level tested universe and all named immune candidates meeting the original aims/methods definition in accessible main text and supplements, regardless of significance. Retain reported feature counts and linked accessible supplements rather than inventing thousands of unreported rows. Record any candidate-selection rule and unavailable supplementary data. Preserve result units and avoid selecting only significant features. Unreported effect sizes are NR. Figure-derived extraction must identify the source figure/panel, reading or digitization method, software/settings, units and uncertainty handling; two reviewers independently check the source and values and retain their extractions and reconciliation. Estimated values are not presented as directly reported data. Where reliable extraction is impossible, use NR or UNCLEAR with the reason.

Participant count, biological sample count, cell count and technical repeats are different fields. The donor is the independent unit for single-cell inference; cells do not replace participants. Model splits are assessed at the participant/cohort level, including feature selection inside training folds. Raw individual change scores without measurement/within-person noise assessment do not establish reproducible responder status.

G.3 Time representation

Exact exercise-to-sampling times are primary. Optional display bins are: pre-exercise baseline; matched control time; during exercise (ancillary); 0 to <30 minutes; 30 minutes to <3 hours; 3 to

<24 hours; 24 to 72 hours inclusive; >72 hours; and NR. Bins up to and including 24 to 72 hours lie within the organizing window; >72 hours is outside-window recovery. Bins are mutually exclusive display categories, not eligibility thresholds. Unknown timing is not imputed to a bin; uncertainty about exercise linkage may prevent main-map eligibility. Symptom/infection follow-up uses a separate time axis, with marker measurement before, concurrent with or after onset recorded explicitly.

H. Critical appraisal

Critical appraisal is prespecified to explain evidence limitations, not to create a clinical recommendation. Two reviewers independently apply tools matched to actual design, with discussion and adjudication. Current JBI tools are planned for randomized trials (2023 revision), quasi-experimental studies (2024), cohorts (2025), analytical cross-sectional studies (2026), and other eligible designs using the applicable current checklist [12]. Single-group pre/post studies are not treated as randomized trials. Crossover/repeated-measure limitations are charted explicitly. True diagnostic-accuracy questions use the current JBI diagnostic-accuracy tool. True individual-level future clinical-outcome or immune-response prediction-model development/validation studies additionally use PROBAST+AI (2025) [13]; a statistical model, molecular discovery classifier or simple association alone does not trigger that tool.

The exact official forms, versions, access dates and checksums will be archived before appraisal. Native response categories, domains and supporting source passages are retained. We do not assume all JBI forms have identical item structure or force them into a common score. The manual in `05_extraction/critical_appraisal_manual.md` governs design selection and overlap.

Assay/omics reporting and the seven precision domains are recorded separately from design appraisal. They include analytical reliability, pre-analytical handling, normalization, cell composition, repeated-measure analysis, multiplicity and independent validation. These custom descriptors are not a validated risk-of-bias tool. There is no quality sum, quality-threshold exclusion, GRADE certainty assessment or numerical readiness ranking; domain evidence enters only the validation-readiness map as defined in A. Appraisal findings contextualize the maps, including unresolved reporting limitations.

I. Synthesis and planned outputs

Synthesis is descriptive, stratified and narrative. Independent cohorts are the primary count for evidence density; report counts and distinct studies are also reported with explicit linkage. Platform features, cells, assays or time points are never used as proxies for evidence strength. Denominators are specific to each question and include applicable/NR/UNCLEAR information as appropriate. A/B overlap is disclosed, and the two stream totals are not simply added.

Three primary evidence maps are planned:

1. **Open-window time by immune compartment and marker family:** Exact sampling coverage and optional time bins within and beyond the organizing window, by immune compartment or matrix, marker family, exercise context and population stratum.
2. **Per-marker validation-readiness map:** Marker or marker family by the seven precision domains, with conventional and omics candidates distinguishable; the descriptive profile defined in A, not a score or ranking.
3. **Cohort-level validation links:** Technical confirmation, repeated bouts, independent participants/cohorts, directly measured function, tested associations and clinical/prediction outcomes, with explicitly labelled gaps.

The PRISMA flow is an additional process figure. Main tables describe study/cohort context, measurement/timing and limitations, individualization/validation evidence, and differences from

adjacent reviews. Core A, support B and the frontier preprint register are visibly separated. Background sources do not enter original-study denominators. Counts of studies reporting a direction may be descriptive but are not a vote-counting estimate of effect or truth. Null, contradictory and unreported findings remain visible.

We will not pool effects, derive a diagnostic threshold, validate a marker, recommend return-to-training decisions, conclude that a candidate predicts infection from group differences alone, or adjudicate whether an open window exists. Discovery of a potentially homogeneous subset will be documented as a future focused-review opportunity. Any later decision to add an effect synthesis requires an explicit protocol amendment and appropriate methods before analysis. The manuscript will retain the scoping methodology regardless of the submission portal's article-category label [14].

J. Registration, archiving, governance and transparency

The protocol is archived publicly rather than registered in a review registry. After the research team freezes version 3.0, the public GitHub repository will be tagged and a GitHub Release archived automatically by Zenodo, which assigns a dated, immutable version DOI and a concept DOI for the record series; the DOI is assigned on release and cannot be stated earlier. The release precedes the 50-record screening pilot and the 10-report charting pilot, and formal five-database searching starts only after the archived release exists. If the pilots change a rule, version 3.1 is released with justification, as a new tag and version DOI under the same concept DOI, before formal searching. PROSPERO does not accept scoping reviews, and OSF Registries is no longer the planned route. At the date of this draft no release, DOI or registration of any kind exists, and the project has never been registered in any registry.

Pre-archive knowledge is disclosed rather than denied. An internal protocol (version 1.0, January 2026) with its own literature collision assessments, public reconnaissance and source verification on 2 October 2026 (Codex), and a collision re-check of the narrowed framing against published and registered reviews on 4 October 2026 (Claude Code; 02_preliminary/collision_recheck_2026-10-04.md) all precede the archive, and the two pilots are planned stages that will also precede formal searching. Their dates, scope and influence on decisions are recorded in the archive release record and the amendment log. The work is not described as wholly prospective from before any knowledge of the literature.

Substantive amendments record the previous rule, new rule, reason, date, work stage, information already known, authorizing team decision and need to revisit earlier records; after the archive release, each amendment also produces a new tagged Release and version DOI. Raw search exports and independent decisions remain immutable. The canonical protocol and execution appendices are versioned together.

This review uses published or lawfully accessible material and does not recruit participants or intervene in exercise. Institutional requirements for ethical review/exemption will be checked and recorded; no approval or exemption number is invented. The GitHub repository and Zenodo archive hold the protocol, manuals, templates, search strategies, dictionaries, decision metadata and shareable synthesis data under CC BY 4.0 for documents and MIT for scripts; copyrighted full texts and raw database exports are not redistributed contrary to their terms. Author names, affiliations, funding and competing interests are pending human declarations.

AI has supported public-source reconnaissance, evidence organization, consistency checking and drafting in this project: Codex desktop with delegated agents on 2–3 October 2026, and Claude Code (Anthropic) on 4 October 2026 for the collision re-check and this v3.0 revision. Tool names, versions, models, sources, dates, tasks, prompt/output references and human verification are logged in 01_protocol/ai_use_log.json, and the manuscript will disclose tool name, version, model and source in both Methods and Acknowledgments [14]. AI is not an author, human

independent reviewer or adjudicator. Named humans remain responsible for eligibility, data, appraisal, citations and conclusions. No AI-produced citation or numerical claim is accepted without source verification. Final disclosure follows the then-current journal and institutional requirements.

K. Roles and execution plan

Roles to name are: principal investigator; reviewers A and B; search lead; independent PRESS reviewer; adjudicator; data/version manager; and analysis/writing lead. A second human reviewer and access to all five databases/full texts are available. Other roles may overlap where independence is preserved, but an adjudicator must be named before formal screening. Funding and competing interests cannot be inferred from the absence of information.

Screening, extraction and appraisal use Excel workbooks generated from the JSON templates in `04_screening` and `05_extraction` by `scripts/build_workbooks.py`: a read-only record master, one locked decision workbook per reviewer, dropdown-controlled codes, and identifier and gene-name columns stored as text. Each reviewer's workbook is locked at completion and its SHA-256 hash committed to git before the two files are merged; `scripts/merge_screening.py` performs the deduplication checks, merge, agreement, kappa and PRISMA counts, and CSV exports are versioned in git. Rayyan, Covidence or similar platforms are not used. A reference manager with exportable identifiers is still to be chosen; software and versions are recorded before execution.

Preparation is estimated at approximately 2–3 working weeks, depending on team availability: 3–5 working days to verify platforms and search strategies and to generate the workbooks; 2–3 working days to resolve team comments, freeze version 3.0 and publish the archived release; and 5–7 working days for reviewer training and the pilot screening and charting, with parallel work where feasible. If the pilots change a rule, a conditional re-release follows before formal searching. These are planning estimates, not a promised completion date. Full-review effort will be estimated after the 50-record screening pilot and 10-report charting pilot, using actual yields and per-record effort. A provisional budget should distinguish two-person screening/extraction from single-person work and allow adjudication and update searches.

The local execution plan and archive release record specify stage gates and deliverables. Formal stages remain pending until the required human work is completed; completing this document does not certify PRESS review, calibration or archiving.

L. Version history and decision status

Version 1.0 (January 2026) and associated build artifacts are preserved in the dated archive. Version 2.0 (2 October 2026) replaced the open-window-led title with candidate biomarkers for post-exercise precision monitoring; defined seven independent precision domains; separated acute A from strictly time-linked repeated-bout B; retained immediate samples without a 72-hour ceiling; expanded searching to inception without a language filter; removed mandatory precision/biomarker/open-window terms; added cohort linkage, complete outcome/validation fields, design-appropriate descriptive appraisal and AI transparency; and removed unsupported novelty claims.

Version 3.0 (this draft, 4 October 2026) follows the collision re-check of the same date. It re-adopts the open window as the organizing construct that defines the 0–72 hour post-exercise window, the sampling time points and the marker families, without adjudicating whether it exists; defines the per-marker validation-readiness map as the deliverable; removes any exercise intensity, duration or mode threshold; restricts the population to adults aged 18 years or older, with older adults as a subgroup; retains Support B as the cross-bout monitoring layer; adds the author open-window interpretation tag, marker family and window-position fields; replaces OSF registration with a public GitHub repository and a Zenodo-archived Release, to be published after team freeze and

before the pilots; specifies Excel workbooks generated from the JSON templates with scripted merging and agreement; adopts CC BY 4.0 for documents and MIT for scripts; and adds the Frontiers in Immunology submission requirements (PRISMA-ScR checklist and flow diagram, AI-use disclosure in Methods and Acknowledgments). Amendment PRE-003 (4 October 2026, decision by A; included in the v3.0 release) makes the post-exercise/acute-bout block T a required search concept (E AND (I OR O) AND T) on PubMed validation evidence, with an expanded seed set, a T-free pilot sensitivity run and citation chasing as mitigations, and records the PRESS conditions (no infection-outcome terms; **train*** retained pending the pilot test). The v3.0 date becomes the actual team freeze date at release.

The scope and operational design are ready for team freeze. Still pending are named administrative declarations, verification of the four licensed database platforms, the Zenodo release, human pilots, archived appraisal forms and local software configuration. Subsequent formal searches, screening, extraction and synthesis are distinct execution stages and have not been performed.

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